Team 4 | CS 4342 Database Systems | Assignment 3

Desert BLoom Market



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Project Prompt



A local supermarket chain has hired you to keep track of their branches, suppliers, staff, and items.

We are interested in storing the branch name, address, and the email, phone number and full name of the branch manager. Branches sell items which have a unique code, a name, categories (for example, a lawn chair can be found in the outdoors and in the furniture sections), description, and price. Items are provided by the suppliers, we care about the country they are manufacture in, owner, company name, and the email and phone number of their sales representative.

Each item can be sold at different branches, has a singular supplier, has a unique barcode, and is stored with a name, description, price, and stock amount.

Staff members work at one or more branches, the system stores their position, date they were hired, date of birth, monthly salary, their full name (first and last), and computes the time that they have worked for the supermarket.

Assumptions

- The entire coding part was done on MySQL.
- Each client can only purchase one product at a time.
- The branch location is included in the store name.
- Each staff member has a unique employee ID.
- We have an automatic system that orders supplies from suppliers.
- The database shall keep track of each branch’s supplier, staff, and items, including the general activity going on per Branch.



Reports

We are interested in obtaining the following reports from the information stored in the database:

• A list of the total amount paid in salaries for each branch, ranked in descending order.

• A list of the branch managers, next to the contact information of the sales representatives who provide the items for their branch.

• A simple search function that returns the names of branches where a product is sold at, note that it is assumed that the item is always in stock.

• A full list of branches and the names of their employees.



Our Entities

Supplier



Item



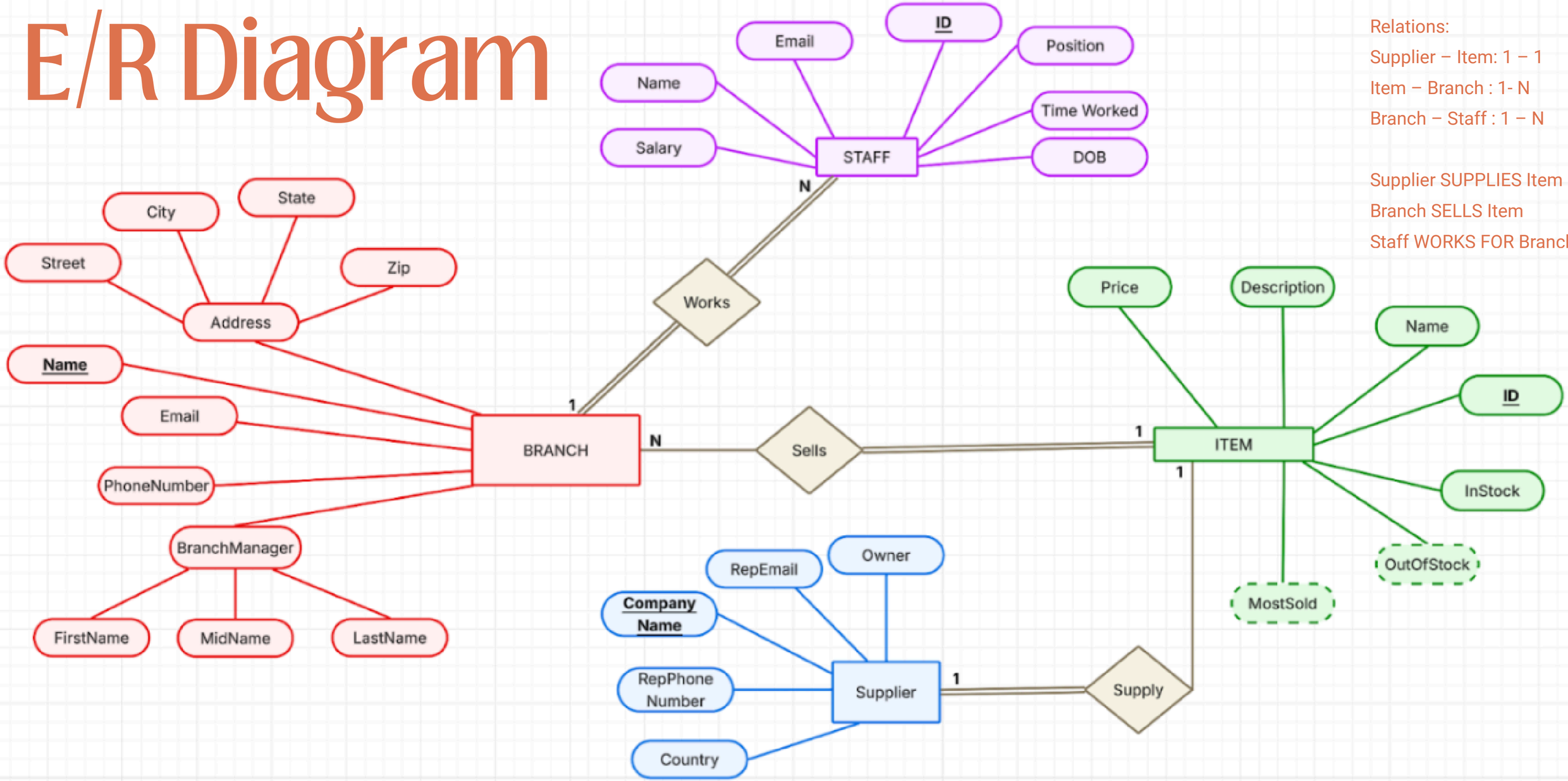
Staff



Branch



E/R Diagram



Relations:

Supplier – Item: 1 – 1

Item – Branch : 1- N

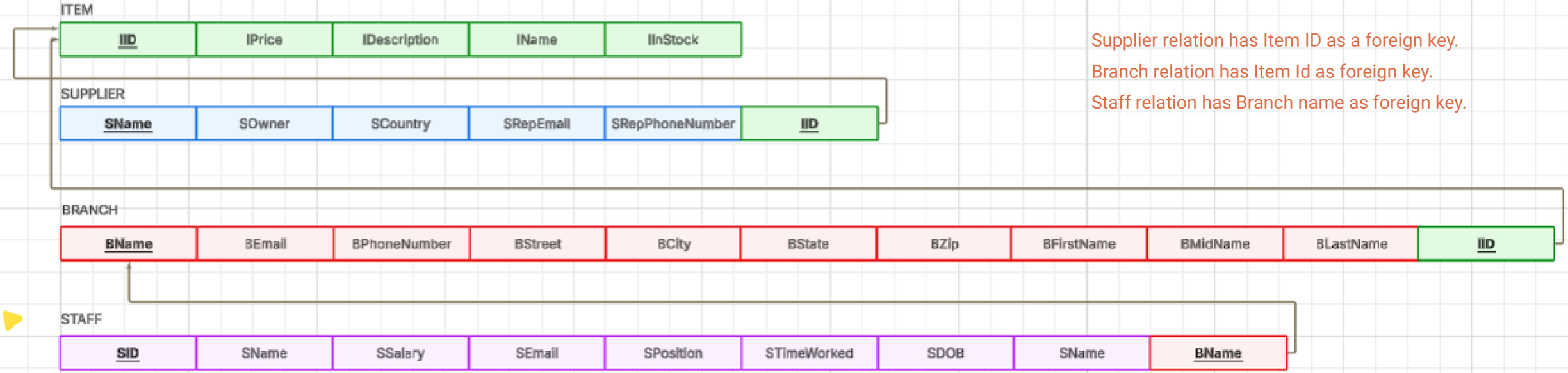
Branch – Staff : 1 – N

Supplier SUPPLIES Item

Branch SELLS Item

Staff WORKS FOR Branch

Relational Model



Supplier relation has Item ID as a foreign key.

Branch relation has Item Id as foreign key.

Staff relation has Branch name as foreign key.

Normalized Schema

Supplier
FD1(trivial):{SName}->{SOwner,SCountry,SRepEmail,SRepPhoneNumber}
Staff
FD1:{SID}->{Sname,SSalary,SEmail,SPosition,STimeWorked,SDOB}
Item
FD1(trivial): {IID} -> {IPrice,IDescripton,IName, IInStock}
Branch
FD1 (trivial): {BName} -> {BEmail, BPhoneNumber, BStreet, BCity, BState, BZip, BFirstName, BMidName, BLastName, IID}
Applies to all:
The relation is in First Normal Form(1NF) since all of the attributes are atomic.
The relation is in Second Normal Form(2NF) since the attributes are fully dependent on only the Primary Key and not any non prime attributes.
The relation is in Third Normal Form(3NF) since there is no transitive dependency between any non-prime key attributes.

SUPPLIER

SName	SOwner	SCountry	SRepEmail	SRepPhoneNumber

BRANCH

BName	BEmail	BPhoneNumber	BStreet	BCity	BState	BZip	BFirstName	BMidName	BLastName

ITEM

IID	IPrice	IDescription	IName	IInStock

STAFF

SID	SName	SSalary	SEmail	SPosition	STimeWorked	SDOB	SName

Supplier Entity



Description: The Supplier entity is used to keep track of Suppliers that supply to the supermarket – specifically, the Name of the Company, the name of the Owner, the contact information of the representative, and the country where items(with unique item ids) are manufactured.

Design Choices: The Company Name has to be unique(acts as a primary key).

Relation with Item: The Supplier table stores the Item Id to keep track of what supplier provides what item to thesupermarket. This is done by having the Item ID attribute referenced as a foreign key.

Implementation in MySQL :

```
MySQL dbserver2.utep.edu:33060+ ssl f25_akl_team4 SQL > DESCRIBE Supplier;
+----------------+--------------+------+-----+---------+-------+
| Field          | Type         | Null | Key | Default | Extra |
+----------------+--------------+------+-----+---------+-------+
| CompanyName    | varchar(30)  | NO   | PRI | NULL    |       |
| RepPhoneNumber | varchar(30)  | YES  |     | NULL    |       |
| Country        | varchar(30)  | YES  |     | NULL    |       |
| Owner          | varchar(30)  | YES  |     | NULL    |       |
| RepEmail       | varchar(30)  | YES  |     | NULL    |       |
| IID            | int unsigned | YES  | MUL | NULL    |       |
+----------------+--------------+------+-----+---------+-------+
```

```
MySQL dbserver2.utep.edu:33060+ ssl f25_akl_team4 SQL > SELECT * FROM Supplier;
+-------------+----------------+---------+---------+-----------------+-----+
| CompanyName | RepPhoneNumber | Country | Owner   | RepEmail        | IID |
+-------------+----------------+---------+---------+-----------------+-----+
| Company A   | 123456789      | China   | Owner A | ownera@gmail.com |   1 |
| Company B   | 3645623789     | China   | Owner B | ownerb@gmail.com |   2 |
| Company C   | 8675349103     | India   | Owner C | ownerc@gmail.com |   3 |
+-------------+----------------+---------+---------+-----------------+-----+
```

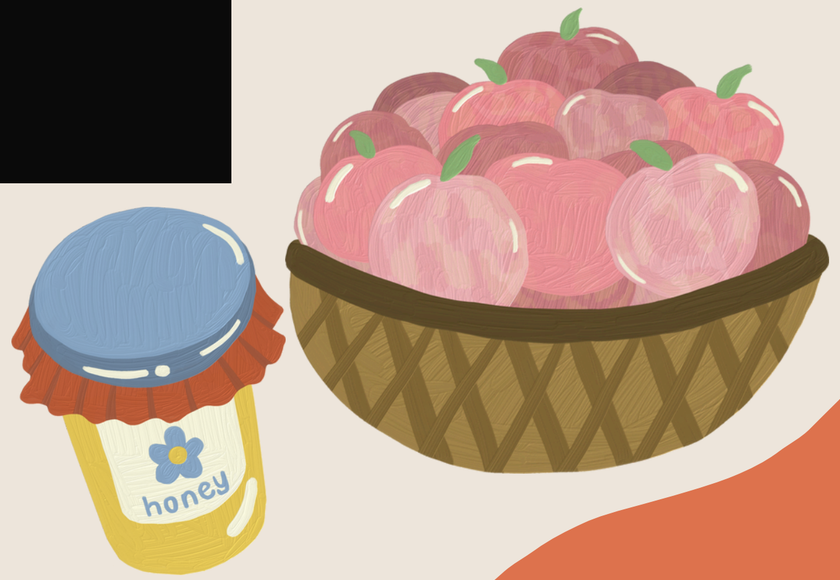
Item Entity

```
MySQL dbserver2.utep.edu:3306 ssl f25_akl_team4 SQL > DESCRIBE Item;
+--------------+---------------+------+-----+---------+----------------+
| Field        | Type          | Null | Key | Default | Extra          |
+--------------+---------------+------+-----+---------+----------------+
| IID          | int unsigned  | NO   | PRI | NULL    | auto_increment |
| IName        | varchar(100)  | NO   |     | NULL    |                |
| IDescription | varchar(225)  | YES  |     | NULL    |                |
| IPrice       | decimal(10,2) | NO   |     | NULL    |                |
| IInStock     | int unsigned  | NO   |     | 0       |                |
+--------------+---------------+------+-----+---------+----------------+
5 rows in set (0.0026 sec)
```

```
MySQL dbserver2.utep.edu:3306 ssl f25_akl_team4 SQL > SELECT * FROM Item;
+-----+--------+------------------------+--------+----------+
| IID | IName  | IDescription           | IPrice | IInStock |
+-----+--------+------------------------+--------+----------+
|   1 | Item A | First item description  |  10.00 |        5 |
|   2 | Item B | Second item description |  15.00 |       12 |
|   3 | Item C | Third item description  |  15.50 |        0 |
+-----+--------+------------------------+--------+----------+
```

- **Description:** The Item entity is used to keep track of the items sold in the supermarket. Each item has an ID, name, description, price, stock amount, and barcode.
- **Design Choices:** The item ID has to be unique (acts as a primary key).
- **Relation with Branch:** The Item table stores the Branch name in a report to keep track of where each item is being sold. This is done by having the BNamereferenced as a foreign key.
- **Implementation in MySQL:**

```
MySQL dbserver2.utep.edu:3306 ssl f25_akl_team4 SQL > SELECT IName, BName FROM Item NATURAL JOIN Branch;
+--------+---------+
| IName  | BName   |
+--------+---------+
| Item A | DBWest  |
| Item B | DBEast  |
| Item C | DBNorth |
+--------+---------+
```



Staff Entity



- **Description:** The Staff entity is used to keep track of the staff members' information that work at every branch.
- **Design Choices:** The staff ID has to be unique (acts as a primary key).
- **Relation with Branch:** The Staff table stores the Branch name to keep track of where the current staff member is working. This is done by having the BName referenced as a foreign key.
- **Implementation in MySQL :**

```
MySQL dbserver2.utep.edu:3306 ssl f25_akl_team4 SQL > DESCRIBE staff;
+-------------+-------------+------+-----+---------+-------+
| Field       | Type        | Null | Key | Default | Extra |
+-------------+-------------+------+-----+---------+-------+
| ID          | int         | NO   | PRI | NULL    |       |
| Name        | varchar(50) | YES  |     | NULL    |       |
| Salary      | varchar(10) | YES  |     | NULL    |       |
| Email       | varchar(20) | YES  |     | NULL    |       |
| Position    | varchar(30) | YES  |     | NULL    |       |
| Time Worked | varchar(40) | YES  |     | NULL    |       |
| DOB         | varchar(10) | YES  |     | NULL    |       |
| BName       | varchar(50) | YES  | MUL | NULL    |       |
+-------------+-------------+------+-----+---------+-------+
```

```
MySQL dbserver2.utep... SQL > SELECT * FROM staff
                    -> ;
+----+---------------+--------+-----------------+--------------+-------------+------------+---------+
| ID | Name          | Salary | Email           | Position     | Time Worked | DOB        | BName   |
+----+---------------+--------+-----------------+--------------+-------------+------------+---------+
|  1 | Carlos Rivera | 55000  | carlos@gmail.com | Cashier      | Full-time   | 1995-03-10 | DBWest  |
|  2 | Ana Torres    | 62000  | ana@gmail.com    | Store Manager | Full-time   | 1990-07-14 | DBEast  |
|  3 | Jose Martinez | 48000  | jose@gmail.com   | Stock Clerk  | Part-time   | 1998-12-22 | DBNorth |
+----+---------------+--------+-----------------+--------------+-------------+------------+---------+
```



Demo

Team 4 Page Link

Login information:

Username: user

Password: user1

OR

Username: admin

Password: admin1

Thank You!

